

Problem (given).

A small country is building an ATC (air traffic control) system for its single airport.

- Every arriving aircraft has a transponder that broadcasts aircraft type & flight data in a high-density packet format.
- The ATC ground station receives, unpacks, and stores the data in an aircraft database.
- A graphics display is built from the stored info and updated every 10 seconds for the controller.
- The system analyzes information to detect dangerous situations.
- The controller can query the database for details about any plane on screen.

Task: Use CRC (Class-Responsibility-Collaboration) to design this system. You are required to produce a set of CRC cards only.

Possible Classes Needed:

- Transponder
 - responsibilities: reads aircraft type and flight data from Aircraft, formats and distributes flight and aircraft data
 - `format_flight_data(flight_code)`: calls `get_flight_data()`, calls `get_aircraft_type()`, formats it to "high-density packet" and returns
 - collaborators: Aircraft, Ground_Receiver
- Aircraft
 - responsibilities: has aircraft type & flight data (aircraft code, flight data, etc...)
 - flight data is an instance variable that's a list of hashmaps of the form `{"flight_code": 000, "time":..., "aircraft_code":...}`
 - `get_flight_data(code)`: iterates through list to find map with `flight_code == code`
 - `get_aircraft_type()`: returns aircraft type
 - collaborators: N/A
- Ground_Receiver
 - Responsibilities: receive broadcast packets from Transponder, detect valid transponder signals, forward received packets for unpacking
 - Collaborators: Transponder, PacketProcessor
- PacketProcessor
 - Responsibilities: unpack high-density data packets, extract aircraft flight information, validate and format data before storage
 - collaborators: Ground_Receiver, Database
- Database
 - responsibilities: stores aircraft flight data, update aircraft information as new data arrives, provide aircraft data for display and queries
 - collaborators: PacketProcessor, Display, Controller, Safety_Analyzer

- Display
 - collaborators: Database, Controller
 - responsibilities: retrieve aircraft info from database, visually display of aircraft positions, update controller every 10s
 - `get_flight_info()`: retrieves most up-to-date info from database
 - `show_display()`: shows most recently retrieved info
 - `update_display()`: calls `get_flight_info()` and `show_display()`
- Safety_Analyzer
 - Responsibilities: analyze aircraft data for dangerous situations, detect conflicts (e.g., proximity, collision risk), alert controller when hazards are detected
 - Collaborators: Database, Controller
 - `assess_danger()`: retrieves info from Database, checks data againsts baselines, if safe: print ("All Systems Stable"), otherwise call `alert_controller()`
 - `alert_controller()`: alerts Controller of specific safety issue
- Controller
 - Responsibilities: queries aircraft details, display warnings and aircraft information, makes controller commands or requests
 - collaborators: Database, Safety_Analyzer, Display
 - `get_details(aircraft_code)`: query the database for details about any plane on screen.

Solution

Class: **Aircraft**

Responsibilities:

- Maintain aircraft identity and type
- Maintain current flight data
- Provide aircraft information when requested

Collaborators (if any):

- Transponder

Assumptions (if any):

- Each aircraft has a transponder installed
- Aircraft flight data is continuously updated

Class: **Transponder**

Responsibilities:

- Collect aircraft type and flight data
- Encode aircraft information into high-density packets
- Broadcast encoded packets to the ground station

Collaborators (if any):

- Aircraft
- Ground Receiver

Assumptions (if any):

- Transponders periodically broadcast aircraft data
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Class: **Ground Receiver**

Responsibilities:

- Receive broadcast packets from aircraft transponders
- Detect and validate incoming signals
- Forward received packets for processing

Collaborators (if any):

- Transponder
- Packet Processor

Assumptions (if any):

- The receiver can handle multiple incoming aircraft signals
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Class: **Packet Processor**

Responsibilities:

- Unpack high-density data packets
- Extract aircraft flight information
- Validate and format data before storage

Collaborators (if any):

- Ground Receiver
- Database

Assumptions (if any):

- Packets follow a known high-density format
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Class: **Database**

Responsibilities:

- Store aircraft flight data
- Update aircraft records when new data arrives
- Provide aircraft data for system components

Collaborators (if any):

- Packet Processor
- Display
- Controller
- Safety Analyzer

Assumptions (if any):

- Each aircraft record can be uniquely identified
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Class: **Display**

Responsibilities:

- Retrieve aircraft information from the database
- Construct a graphical display of aircraft positions
- Update the controller's display every 10 seconds

Collaborators (if any):

- Database
- Controller

Assumptions (if any):

- The display refresh cycle is fixed at 10 seconds
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Class: **Safety Analyzer**

Responsibilities:

- Analyze aircraft data for dangerous situations
- Detect potential aircraft conflicts
- Notify the controller when hazards are detected

Collaborators (if any):

- Database
- Controller

Assumptions (if any):

- Safety rules exist for determining dangerous situations
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Class: **Controller**

Responsibilities:

- Query aircraft information
- Request detailed information about aircraft on screen

- Receive safety alerts and warnings

Collaborators (if any):

- Database
- Display
- Safety Analyzer

Assumptions (if any):

- The controller interacts with the system through the display interface